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Image-Based & Sequencing-Based Methods with AI-Assisted Workflows

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PARTS

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Computational Biology & Spatial Genomics

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Spatial Transcriptomics & Bioinformatics

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About the Authors

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Dr. Madhu Sudhana Saddala is a computational biologist and spatial genomics researcher with extensive expertise in single-cell and spatial transcriptomics data analysis. His research focuses on developing and applying computational methods to decipher the spatial organisation of gene expression in complex tissues, with particular emphasis on the brain and tumour microenvironments. Dr. Saddala has contributed to methodological advances in spatially variable gene detection, cell-type deconvolution, and AI-assisted bioinformatics workflows. He is a strong advocate for open science and reproducible research practices.

Research interests: Spatial transcriptomics, single-cell genomics, neural circuit mapping, computational method development, AI-assisted biological discovery.

Preeti Sharma

Preeti Sharma is a bioinformatician and spatial transcriptomics specialist with deep expertise in image-based spatial platforms including Xenium, MERFISH, and CosMx. Her work centres on the development of integrated analysis pipelines that bridge molecular profiling with tissue morphology. She has extensive experience in tumour microenvironment characterisation, ligand-receptor interaction analysis, and multi-modal spatial data integration. Preeti is committed to making spatial transcriptomics analysis accessible to researchers across biological disciplines through clear documentation and educational resources.

Research interests: Image-based spatial transcriptomics, tumour microenvironment, cell-cell communication, multi-omic data integration, computational pipeline development.

Foreword

The spatial revolution in genomics has arrived faster than almost anyone anticipated. When the first spatial transcriptomics papers appeared in the mid-2010s, they were proof-of-concept demonstrations. Within a decade, the field produced commercial platforms offering single-molecule resolution, FFPE-compatible protocols for clinical archives, and multi-modal data types simultaneously capturing chromatin accessibility and gene expression at defined tissue positions.

This textbook grew out of a practical need. When we began working with spatial transcriptomics data, we found that the computational analysis landscape was fragmented: excellent vignettes for individual packages existed, but there was no single resource that took a researcher from raw data all the way through to publication-quality results, covering both major technology families and both Python and R ecosystems in depth.

The book is deliberately practical. Every chapter builds toward working code you can run on real, publicly available data. Where we have made analytical choices — threshold selection, parameter tuning, method comparison — we explain the reasoning so you can adapt those choices to your own data.

We have also treated AI-assisted analysis as a first-class topic. Tools like GitHub Copilot and large language models have genuinely changed how bioinformatics code is written and debugged. A dedicated section on prompt engineering for spatial analysis workflows reflects our belief that this is now a core competency for computational biologists.

We hope this book serves as a useful companion throughout your spatial transcriptomics journey — from first data loading to the figures in your manuscript.

— *Dr. Madhu Sudhana Saddala & Preeti Sharma*

Table of Contents

Part I — Foundations

- 01 Introduction to Spatial Transcriptomics
- 02 Data Formats & the Computational Ecosystem
- 03 Setting Up Reproducible Environments

Part II — Sequencing-Based Spatial Transcriptomics

- 04 Loading & Quality Control: Visium & Slide-seq
- 05 Normalisation & HVG Selection
- 06 Dimensionality Reduction & Clustering
- 07 Spatially Variable Gene Detection
- 08 Cell-Type Deconvolution
- 09 Trajectory & Pseudotime Analysis in Space

Part III — Image-Based Spatial Transcriptomics

- 10 Loading Xenium & CosMx Subcellular Data
- 11 Cell Segmentation & Quality Control
- 12 Single-Cell Resolution Clustering
- 13 Spatial Statistics & Neighbourhood Analysis
- 14 Ligand-Receptor Interaction Analysis
- 15 Tissue Niche & Microenvironment Analysis

Part IV — Integration & Multi-Modal Analysis

- 16 Integrating Spatial + scRNA-seq References
- 17 Multi-Sample & Multi-Slide Integration
- 18 Joint Image + RNA Analysis
- 19 Spatial Multi-Omics

Part V — AI & Codex-Assisted Analysis

- 20 Using Codex for Spatial Analysis Workflows
- 21 Foundation Models for Spatial Data
- 22 AI-Assisted Figure Generation
- 23 Automated Reporting with Quarto

Part VI — Case Studies & Workflows

- 24 Case Study: Mouse Brain Atlas (Visium)
- 25 Case Study: Human Breast Cancer (Xenium)
- 26 Case Study: Intestinal Organoids (MERFISH)
- 27 Case Study: Spatial Multi-Omics (Multiome)
- 28 Best Practices & Pitfalls

Appendices

- Appendix A Python Package Reference
- Appendix B R Package Reference
- Appendix C Example Datasets & Downloads